

INTELLECTUAL PROPERTY DISCLOSURE DOCUMENT

NotiFetch - Cross-Platform Delivery Notification Aggregator
Confidential - Prepared for Institutional Review

1. Overview

This document formally discloses the intellectual property assets associated with NotiFetch, a cross-platform notification aggregation platform engineered for gig economy delivery partners. NotiFetch captures, parses, and consolidates order notifications from over 50 delivery applications into a single unified real-time feed, employing proprietary technical methods that constitute patentable inventions. This disclosure is submitted in support of the startup funding application to Sapthagiri NPS University and serves as a formal record of the intellectual property created by the founding team.

Detail	Information
Product Name	NotiFetch
Enterprise	NotiFetch (Proprietorship - Udyam Registered)
Founder	Yashas K. Gangatkar
Product URL	https://notifetch.app
Platform Coverage	50+ delivery partner applications across 8 global regions
Technology Stack	Next.js 16, Kotlin (Android), Prisma, PostgreSQL, Firebase, Razorpay, Stripe
Date of Disclosure	June 2026

2. Invention I: Multi-Language, Multi-Currency Notification Parsing Engine

2.1 Technical Description

The core parsing engine is a proprietary system that extracts structured, actionable data from unstructured notification text generated by over 50 distinct delivery platform applications. Each delivery platform dispatches notifications in a unique format with varying text structures, language conventions, currency notations, and data field arrangements. The engine employs a hierarchical regex-based parsing architecture with platform-specific pattern libraries, enabling it to identify and extract the following data fields from any supported platform notification:

Data Field	Example Extraction	Technical Challenge
Order Value	Rs. 150, \$12.50, R\$25.00	28 currency formats with varying symbols, comma/decimal conventions
Pickup Location	"Pick up at Koramangala Hub"	14 languages, varied address formats, embedded within free text
Drop-off Location	"Deliver to HSR Layout"	Distinguish from pickup using context-aware keyword detection

Distance	3.5 km, 2.2 mi	Mixed unit systems (metric/imperial) across regions
Notification Category	NEW_ORDER, EARNINGS, COMPLETED	7 categories inferred from unlabelled notification text

2.2 Novelty and Non-Obviousness

The novelty of this invention lies in the unified parsing architecture that handles 50+ distinct notification formats through a single extensible framework, rather than requiring separate parsers or API integrations for each platform. The non-obvious aspect is the hierarchical pattern-matching approach that first identifies the source platform from the notification package name, then applies a platform-specific regex library tuned for that platform's notification format, while simultaneously handling multi-language text (English, Hindi, Japanese, Portuguese, Spanish, Arabic, Indonesian, Korean, and 6 additional languages) and multi-currency value extraction (28 currencies) within the same parsing pass. No existing system provides this level of cross-platform, cross-lingual, cross-currency structured data extraction from passive notification interception.

2.3 Potential Patent Claims

1. A method for extracting structured delivery order data from unstructured mobile notification text, comprising: identifying the source application; selecting a platform-specific parsing pattern library; applying multi-language regex patterns to extract order value, pickup location, drop-off location, distance, and notification category; and normalizing extracted values across multiple currency and language formats into a unified data structure.
2. The method of claim 1, wherein the platform identification is performed via Android package name matching against a verified database of 50+ delivery platform identifiers.
3. The method of claim 1, wherein currency normalization handles 28 distinct currency formats including symbols, comma conventions, and decimal placement variations.

3. Invention II: Zero Credentials Architecture for Cross-Platform Data Acquisition

3.1 Technical Description

The Zero Credentials Architecture is a novel system design that enables cross-platform data aggregation from delivery partner applications without requiring any form of authentication credentials - no API keys, no OAuth tokens, no platform login credentials, and no scraping of authenticated web sessions. Instead, the architecture leverages Android's built-in NotificationListenerService as a passive, user-permissioned data acquisition layer. This system service, accessible only after explicit user authorization through Android's security settings, provides a real-time stream of all notifications displayed on the user's device. NotiFetch's innovation is in recognizing that this system service can serve as a legally compliant, platform-agnostic data source for delivery order aggregation - a use case that the API was not designed for, but which it enables without violating any platform Terms of Service.

3.2 System Architecture

Layer	Component	Function
Acquisition	Android NotificationListenerService	Passive capture of all device notifications with explicit user permission

Filtering	Package Name Matcher	Identifies delivery platform from notification source package (50+ verified package names)
Parsing	Multi-Language Regex Engine (Invention I)	Extracts structured data from notification text
Storage	Room Database (Offline-First)	Local SQLite storage with sync tracking for offline operation
Sync	WorkManager + Firebase	Periodic background sync (15-min intervals) with real-time FCM push
Presentation	Next.js Web Dashboard + Kotlin App	Unified feed, earnings analytics, smart order ranking

3.3 Novelty and Non-Obviousness

The novelty is the repurposing of Android's NotificationListenerService - originally designed for smart reply and notification management - as a credential-free, cross-platform data acquisition layer for commercial delivery order aggregation. The non-obvious aspect is the architectural insight that notification interception can replace traditional API integration entirely, eliminating dependency on platform partnerships, avoiding API rate limitations and authentication complexity, and circumventing the fundamental chicken-and-egg problem of requiring platform cooperation to build a cross-platform tool. This approach also ensures complete legal compliance, as the data accessed is already visible to the user on their device, and the permission model is governed by Android's own security framework.

3.4 Potential Patent Claims

1. A system for cross-platform delivery notification aggregation without platform credentials, comprising: a passive notification listener operating within the Android NotificationListenerService framework; a package name filter identifying delivery platform sources; and a parsing engine extracting structured order data from notification text.
2. The system of claim 1, wherein no API keys, OAuth tokens, or platform login credentials are required for data acquisition from any supported delivery platform.
3. The system of claim 1, wherein the offline-first architecture ensures continuous operation through local Room database storage with periodic WorkManager-based synchronization.

4. Invention III: AI-Powered Smart Order Ranking Algorithm

4.1 Technical Description

The Smart Order Ranking algorithm is a proprietary machine learning model that evaluates and ranks incoming delivery order notifications in real-time based on multiple weighted factors to maximize the delivery partner's earnings efficiency. The algorithm processes each parsed notification (output from Invention I) and computes a composite score based on the following factors:

Factor	Weight	Description
Earnings Potential	35%	Order value relative to regional average, accounting for surge pricing and bonus incentives

Distance Efficiency	25%	Ratio of order value to pickup-to-dropoff distance, optimizing earnings per kilometer
Time Sensitivity	20%	Estimated time-to-acceptance window derived from historical cancellation patterns
Historical Acceptance	15%	Partner's past acceptance rate for similar orders from the same platform
Platform Reliability	5%	Historical order completion rate and payment reliability of the source platform

4.2 Novelty and Non-Obviousness

The novelty lies in the application of multi-factor scoring to delivery order notifications aggregated from multiple platforms - a scenario that does not exist in single-platform environments where only one source of orders is available. The non-obvious aspect is the cross-platform normalization required to compare orders from different platforms that use different pricing models, distance calculations, and incentive structures. The algorithm must normalize these heterogeneous data points into a single comparable score, adjusting for platform-specific biases in real-time. No existing system performs cross-platform order ranking because no existing system aggregates orders from multiple platforms into a single feed.

4.3 Potential Patent Claims

1. A method for ranking delivery order notifications from multiple heterogeneous platforms, comprising: parsing structured order data from platform-specific notification formats; computing a composite score based on earnings potential, distance efficiency, time sensitivity, historical acceptance patterns, and platform reliability; normalizing scores across platforms with differing pricing models; and presenting ranked orders in a unified feed.
2. The method of claim 1, wherein the composite scoring weights are dynamically adjusted based on the delivery partner's historical earnings data and platform preference patterns.

5. Prior Art Analysis

A comprehensive prior art search confirms that no existing patent or published application describes a system combining passive notification interception with multi-language parsing and cross-platform order ranking for the gig delivery segment. The following table summarizes the closest prior art:

Prior Art	What It Does	How NotiFetch Differs
Generic notification managers (e.g., Notification History apps)	Log all notifications on device for user review	No delivery-specific parsing, no structured data extraction, no earnings analytics, no cross-platform intelligence
Platform SDK integrations (e.g., Uber API, Swiggy Partner API)	Access single-platform order data via authenticated APIs	Single-platform only, requires API credentials and platform partnership, not cross-platform
Web scraping tools for delivery platforms	Extract order data by scraping platform web interfaces	Violates Terms of Service, requires authentication credentials, breaks on UI changes, legally indefensible

Gig worker earnings trackers (e.g., Solo, Gridwise)	Track earnings across gig platforms using manual input or limited API access	No real-time notification aggregation, no passive data capture, limited to earnings tracking without order intelligence or smart ranking
---	--	--

6. Commercial Status and IP Protection Plan

All three inventions described herein are currently embodied in the NotiFetch product, which is deployed and commercially operational. The source code implementing these inventions is maintained as a proprietary codebase with access restricted to the founding team. The following IP protection strategy is proposed:

Action	Timeline	Status
Provisional Patent Application (Invention I - Parsing Engine)	Within 6 months of funding	Pending institutional support
Provisional Patent Application (Invention II - Zero Credentials Architecture)	Within 6 months of funding	Pending institutional support
Provisional Patent Application (Invention III - Smart Order Ranking)	Within 12 months of funding	Pending institutional support
Source Code Trade Secret Protection	Ongoing	Active - restricted access
Trademark Registration (NotiFetch)	Within 3 months	In progress

7. Declaration

I, **Yashas K. Gangatkar**, founder and proprietor of NotiFetch, hereby declare that the inventions described in this document are original works created by me. To the best of my knowledge, these inventions do not infringe upon any existing patents, trade secrets, or intellectual property rights of any third party. The technical descriptions, novelty claims, and potential patent claims presented herein are accurate and truthful representations of the proprietary technology embodied in the NotiFetch product. I understand that this disclosure document serves as a formal record of the intellectual property creation date and scope, and may be used as supporting evidence in future patent applications.

Disclosed By: Yashas K. Gangatkar

Designation: Founder and Proprietor, NotiFetch

Signature: _____

Date: _____

Place: Bengaluru, Karnataka

This document is confidential and intended solely for institutional review in connection with the startup funding application at Saphthagiri NPS University. Unauthorized reproduction or distribution is prohibited.